

3 Dynamics

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Newton's second law relates resultant force to the rate of change of momentum: $F = \Delta p / \Delta t$, reducing to ma for constant mass. Momentum is a vector and is conserved in an isolated interaction. During a collision, increasing contact time reduces average force for a fixed change in momentum.

Worked example: follow the reasoning

A 0.20 kg ball reverses velocity from +8 to -6 m s^{-1} . $\Delta p = 0.20(-6 - 8) = -2.8 \text{ kg m s}^{-1}$; over 0.040 s the average force is -70 N .

Quick check · multiple choice

A For an isolated collision, which total is conserved? [1]

A Kinetic energy always

B Temperature

C Momentum

D Speed of each body

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A 0.6 kg cart at 4 m s^{-1} collides and sticks to a stationary 0.4 kg cart. Find their common velocity and state the condition for conservation. [3]

Final answer: _____

Answers and explanation

Quick check: C. Momentum

Question 1 [3 marks]

Initial momentum $0.6 \times 4 = 2.4 \text{ kg m s}^{-1}$ [1]; common velocity $2.4/(0.6+0.4) = 2.4 \text{ m s}^{-1}$ [1]; negligible external resultant impulse [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.